

AI & COMPUTER VISION SPECIES CLASSIFIER

Project Overview

AI-powered deep-learning web application that identifies snake species from an uploaded image.

Problem Statement:	Rapidly identifying venomous versus non-venomous snake species in nature is difficult for non-experts, creating severe safety risks.
Solution Implemented:	Built an end-to-end computer vision web app combining OpenCV image preprocessing with a trained TensorFlow CNN model.
My Technical Role:	AI / ML Engineer & Web Developer

Main Features & Technical Stack

Core Features:	Technologies Used:
• Drag-and-drop image upload with real-time browser preview • Automated image pre-processing and color channel normalization • Deep CNN multi-class species classification • Venom risk severity warnings & safety guidelines	Python, Flask, TensorFlow, Keras, OpenCV, NumPy, HTML, CSS, JavaScript
	Architecture Flow:
	Web UI → Flask Backend → OpenCV Preprocessor → TensorFlow CNN → Vercel Host

Technical Challenge & Results

Key Engineering Challenge: Handled wild image lighting variations via heavy data augmentation (rotations, brightness scaling, color jitter).
Validated Results: Instantaneous species prediction with clear safety risk indicators.

Live Demo: <https://snake-verse-five.vercel.app> | **GitHub:** <https://github.com/prashant-singh-78> | **Portfolio:** prashant08.vercel.app